

BARSHATAS, Kazakhstan

WMO: 364980

Lat: 48.162N

Lon: 78.675E

Elev: 643

StdP: 93.84

Time Zone: 6.00 (E06)

Period: 82-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.2	-25.9	-31.4	0.2	-27.8	-29.1	0.3	-25.3	12.5	-13.8	10.5	-10.5	2.3	50	0.570

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.4	32.3	16.6	30.7	16.1	29.1	15.6	18.1	27.9	17.3	27.3	16.6	26.5	2.7	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.0	11.5	20.3	13.9	10.7	20.6	12.6	9.8	20.6	53.8	28.0	51.2	27.4	49.0	26.3	22.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.7	9.2	7.3	-32.4	35.8	4.4	1.2	-35.5	36.7	-38.1	37.4	-40.6	38.1	-43.8	39.0		
(5)			-31.5	20.2	3.0	1.2	-33.6	21.0	-35.4	21.7	-37.0	22.4	-39.2	23.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.1	-14.0	-13.8	-6.0	6.3	13.1	19.1	21.4	19.2	13.2	5.2	-4.3	-11.6		
	DBStd	13.85	5.44	5.52	6.23	5.31	4.88	4.10	3.35	3.61	4.56	5.11	6.28	6.17		
	HDD10.0	3352	743	667	498	133	27	1	0	1	21	163	428	670		
	HDD18.3	5450	1002	901	756	362	172	38	11	33	163	407	678	928		
	CDD10.0	1189	0	0	1	21	122	276	354	286	116	15	0	0		
	CDD18.3	244	0	0	0	0	8	63	106	60	7	0	0	0		
	CDH23.3	2921	0	0	0	6	132	754	1169	716	139	5	0	0		
	CDH26.7	992	0	0	0	1	26	253	437	246	29	0	0	0		
(14) Wind	WSAvg	2.1	1.8	2.0	2.1	2.5	2.4	2.4	2.2	2.3	2.1	1.9	1.7	1.7		
(15) Precipitation	PrecAvg	207	15	14	11	16	18	19	39	16	6	11	20	18		
	PrecMax	314	31	33	27	39	39	39	115	46	20	34	47	35		
	PrecMin	140	3	5	3	0	2	1	6	0	0	0	2	5		
	PrecStd	49	8	8	8	11	10	9	31	14	5	9	12	9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.0	0.6	14.7	24.3	29.5	33.3	35.7	33.8	30.1	22.9	10.8	1.6		
		MCWB	-2.7	-0.8	8.4	12.3	14.9	17.4	18.2	16.7	14.9	11.7	5.1	-0.2		
	2%	DB	-3.4	-1.6	8.6	20.4	26.7	31.3	32.8	31.5	26.9	19.4	8.2	0.0		
		MCWB	-4.7	-3.2	4.3	10.7	14.3	16.5	16.8	16.0	13.6	9.7	3.8	-0.9		
	5%	DB	-4.9	-3.4	5.2	18.1	24.5	29.7	30.9	29.6	24.7	16.9	5.7	-1.3		
		MCWB	-6.0	-4.8	2.0	9.6	13.3	15.7	16.5	15.4	12.7	8.5	2.6	-2.3		
	10%	DB	-6.3	-5.3	2.4	15.6	22.3	27.8	29.2	27.5	22.5	14.0	3.3	-3.1		
		MCWB	-7.3	-6.5	0.1	8.3	12.0	15.0	16.3	15.0	11.8	7.2	0.8	-4.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.5	-0.4	8.8	13.7	16.5	18.7	19.6	18.1	15.9	12.2	6.4	0.5		
		MCDB	-1.2	0.5	14.1	22.1	26.4	31.2	30.0	28.1	29.2	22.2	9.5	1.0		
	2%	WB	-4.5	-3.0	4.7	11.3	15.0	17.6	18.5	17.2	14.0	10.2	4.4	-0.8		
		MCDB	-3.5	-1.7	8.4	19.3	24.2	28.3	27.9	27.5	25.1	18.5	7.1	0.0		
	5%	WB	-6.0	-4.7	2.4	10.0	14.0	16.7	17.7	16.3	13.0	8.7	2.7	-2.3		
		MCDB	-5.0	-3.4	5.0	17.1	22.6	27.0	27.2	26.6	23.8	15.9	5.7	-1.3		
	10%	WB	-7.2	-6.5	0.1	8.6	12.9	15.9	17.0	15.6	12.0	7.4	0.9	-4.1		
		MCDB	-6.2	-5.3	2.2	15.0	20.9	25.5	26.7	25.8	21.7	13.7	3.1	-3.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.6	9.8	9.7	11.6	12.6	12.6	12.4	12.8	13.0	10.9	9.2	8.3		
		MCDBR	8.8	9.9	11.0	15.3	15.3	14.6	14.7	15.9	16.2	15.2	11.9	8.3		
		MCWBR	8.0	8.9	7.6	8.3	7.4	5.7	5.2	5.9	7.4	8.4	8.1	7.3		
		MCDBR	8.6	9.6	10.8	14.0	13.7	13.0	12.8	13.7	15.3	14.3	10.7	7.4		
	5% WB	MCWBR	7.9	8.7	7.5	8.3	7.9	6.0	5.3	5.9	7.4	8.4	7.8	6.7		
(40) Clear-Sky Solar Irradiance	taub		0.286	0.316	0.370	0.413	0.411	0.415	0.430	0.387	0.344	0.342	0.322	0.289		
	taud		2.102	2.047	2.071	2.099	2.159	2.217	2.188	2.294	2.370	2.285	2.198	2.130		
	Ebn at Noon		756	816	830	833	851	848	828	852	859	788	708	697		
	Edh at Noon		87	117	136	147	144	137	139	120	100	92	80	75		
(44) All-Sky Solar Radiation	RadAvg		1.78	2.90	4.22	5.24	6.65	6.81	6.41	6.02	4.64	2.77	1.55	1.27		
	RadStd		0.11	0.15	0.27	0.34	0.32	0.40	0.40	0.27	0.30	0.23	0.14	0.14		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page